

ONWARDS AND UPWARDS SS -BULOBA
UGANDA ADVANCED CERTIFICATE OF EDUCATION
END OF TERM ONE EXAMINATION 2026

S.6 PHYSICS (P510/1)

Time: 2 hours 40mins

Instructions:

- 1. Attempt one question in each section (ONLY three items in total.)**
- 2. Use the provided data and support materials to guide your solutions**
- 3. Clearly show all the calculations and assumptions.**

SECTION A: PARTICLES (compulsory)

Item 1 (20 marks)

At **MPODE HIGH SCHOOL**, the school nurse, **Nurse Amina**, began noticing a rising number of students reporting to the sick bay with complaints of eye strain, headaches, and disrupted sleep. She linked the problem to the school's newly installed LED computer monitors, which emit bright blue light at approximately 450 nm — a wavelength associated with photon energies that may interact with biological tissue in the eye. She raised her concern with the Head of Science, **Mr. Banda**, a Physics teacher. Mr. Banda organised a classroom investigation to explore the quantum nature of light, specifically the photoelectric effect, using a sodium metal surface and the LED blue light as the source. He challenged the students to determine whether the photon energy of the blue light exceeds the known biological damage threshold for the human retina, to investigate whether the light could cause electron emission from sodium, and to explain to Nurse Amina why the classical wave theory of light fails entirely to account for what is observed in the experiment.

Support Material Provided:

- Wavelength of blue light $\lambda = 450 \text{ nm}$, Planck's constant $h = 6.63 \times 10^{-34} \text{ Js}$
- Speed of light in vacuum $c = 3.0 \times 10^8 \text{ m s}^{-1}$
- Damage threshold for the human eye = 3.1 eV, Work function of sodium $\phi = 2.28 \text{ eV}$
- $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$

Tasks:

(a) Use the information provided to calculate the energy of a single blue light photon in eV, and use this result to advise whether Mr. Banda should reassure students that the LED monitors pose no risk of permanent damage to their retinas. **[4 marks]**

(b) Investigate whether photoelectric emission will occur when blue light from the school monitors falls on the sodium metal surface, and if so, determine the maximum kinetic energy of

the emitted photoelectrons in joules and the stopping potential required to arrest the fastest electrons. **[6 marks]**

(c) Explain to Nurse Amina and the students, in the context of Mr. Banda's experiment, the TWO key observations of the photoelectric effect that the classical wave theory of light completely fails to account for. **[4 marks]**

(d) Advise the school administration on the symptoms students may develop from prolonged exposure to blue light beyond the damage threshold, and suggest specific, physically justified protective measures that could be implemented in the computer laboratory without disrupting learning. **[6 marks]**

SECTION B (FORCE AND MOTION)

Attempt ANY ONE item from this section.

Item 2 (20 marks)

On the morning of 15 January 2026, a serious accident occurred on the Kampala–Jinja Highway near Namanve Industrial Park. A Toyota Premio saloon car of mass 1 200 kg was travelling at 30 m s^{-1} when the driver, Mr. Ssemanda, applied the brakes hard to avoid a stalled heavy goods vehicle, bringing the car to rest in 5.0 seconds on a dry, level road surface and leaving clear skid marks on the tarmac. Traffic investigator Constable Nabirye arrived at the scene to determine whether Mr. Ssemanda had exceeded the highway speed limit of 100 km h^{-1} . A group of Senior Six students from a nearby school happened to be on a field trip to the industrial park. Their Physics teacher, Mr. Opio, transformed the accident scene into a practical lesson, additionally challenging students to analyse the motion of a ball projected horizontally from the roof of a building overlooking the road, and to determine how engine power governs a car's performance both on the level highway and on the hilly terrain that is common on Ugandan roads.

Support Material Provided:

Mass of car $m = 1\,200 \text{ kg}$, initial speed $u = 30 \text{ m s}^{-1}$, braking time $= 5.0 \text{ s}$

Building height $= 80 \text{ m}$, horizontal projection speed $= 20 \text{ m s}^{-1}$

Engine power $= 60 \text{ kW}$, gradient of incline: $\sin \theta = 0.05$

$g = 9.81 \text{ m s}^{-2}$

Tasks: (a) Help Constable Nabirye reconstruct the accident by determining the deceleration of Mr. Ssemanda's car during braking, the length of skid marks she should have found on the road, and the average braking force exerted on the vehicle. (06)

(b) Analyse the motion of the ball that Mr. Opio projected horizontally from the roof of the 80 m building, determining the time of flight, the horizontal distance from the building where it lands, and the magnitude and direction of its velocity at the moment of impact with the road. (07)

(c) Determine the total resistive force acting on Mr. Ssemanda's car as it travels at a constant 30 m s^{-1} on the level highway, and recalculate the resistive force when the same car climbs the incline at the same speed with the same engine power. (05)

(d) Using the impulse-momentum theorem, explain why modern vehicles are equipped with crumple zones and air bags, relating the physics principles to the protection they provided — or would have provided — to Mr. Ssemanda during the collision. (02)

Item 3 (20 marks)

After an accident incident, Constable Nabirye observed a damaged rescue ladder lying near the heavy goods vehicle. It appeared that a mechanic, Mr. Kato (mass 90 kg), had been using a uniform aluminum ladder (mass 30 kg, length 6.0 m) to inspect the truck's roof. The ladder was leaning at an angle of 60° to the horizontal against the smooth vertical side of the truck. To prevent the ladder from slipping on the oily tarmac, Mr. Kato had attached an elastic bungee cord horizontally between the base of the ladder and a sturdy anchor point at the base of the truck wall. The cord has a natural length of 2.0 m and a spring constant of 200 N m^{-1} . Mr. Kato was standing 4.0 m up the ladder (measured along the ladder) when the system was in equilibrium.

Tasks:

- (a) Determine the extension of the elastic cord and calculate the tension force it exerts on the base of the ladder. (05)
- (b) By taking moments about the base of the ladder, calculate the normal reaction force exerted by the smooth truck wall on the top of the ladder. (07)
- (c) Calculate the magnitude and direction of the frictional force acting at the base of the ladder, and determine the minimum coefficient of static friction required between the ladder and the ground to maintain equilibrium. (06)
- (d) Explain, using physics principles, how the presence of the elastic cord improves the stability of the ladder compared to relying on friction alone. (02)

SECTION C ENERGY

| Attempt ANY ONE item from this section.

Item 4 (20 marks)

The Ndere Cultural Centre in Ntinda, Kampala, hosts weekly live performances showcasing Uganda's traditional music. One evening, master performer Kizza played the endingidi — a one-stringed fiddle — before a large audience. The Centre's sound engineer, Ms. Tendo Akullo, used a digital oscilloscope to analyse the waveforms produced by the instruments to calibrate the public address system. She observed that when Kizza bowed his endingidi string of length 0.80 m, fixed at both ends, it vibrated in its third harmonic at a wave speed of 120 m s^{-1} . Before the show, a visiting musician, Sergeant Ocen, drove his car at 15 m s^{-1} towards a stationary outdoor speaker emitting a continuous test tone of 850 Hz. Both Ms. Akullo and bystanders clearly noticed that the pitch Ocen heard differed from what they heard — a phenomenon she used to explain Doppler-based medical ultrasound to an intern. Backstage, a technician also set up a mechanical demonstration for visiting Senior Six students: a 0.50 kg mass attached to a spring of spring constant 200 N m^{-1} was set oscillating with an amplitude of 0.10 m, illustrating simple harmonic motion.

Support Material Provided:

Endingidi string: length = 0.80 m, wave speed = 120 m s^{-1} , third harmonic mode

Speed of sound in air = 340 m s^{-1}

Spring constant $k = 200 \text{ N m}^{-1}$, oscillating mass $m = 0.50 \text{ kg}$, amplitude $A = 0.10 \text{ m}$

Tasks:

Help Ms. Akullo analyse the standing wave on Kizza's endingidi string by sketching the third harmonic pattern with all nodes and antinodes clearly marked, and calculating the wavelength, frequency of vibration in that mode, and the frequency of the fundamental mode of the same string. (07)

Calculate the frequency of sound that Sergeant Ocen hears as his car approaches the outdoor speaker at 15 m s^{-1} , and describe clearly how the same Doppler principle is applied in hospital ultrasound equipment to measure the speed of blood flow in a patient's artery. (06)

© Analyse the simple harmonic motion of the spring-mass system backstage, calculating the angular frequency, maximum speed of the mass, and the total mechanical energy stored in the system. (05)

Using examples drawn from the musical and physical events at the Ndere Cultural Centre described above, state TWO differences between a progressive (travelling) wave and a stationary (standing) wave. (02)

Item 5(20 marks)

FiberLink Uganda Limited recently won a licence to lay fibre optic cables connecting Kampala to Gulu, Mbarara, and Fort Portal. Project engineer Dr. Ssali briefed his technical team that the cables work by trapping light inside a glass core of refractive index 1.52 through total internal reflection. To demonstrate the principle, he directed a laser ray in air onto a flat glass surface at an angle of incidence of 40° and challenged the team to calculate the refraction and critical angles and to explain, using a diagram, how data pulses travel thousands of kilometres without escaping through the sides of the fibre. In Tororo, the Tororo Cement factory's thermodynamicist, Mr. Muwanga, was modelling the factory kiln's heat cycle using an ideal gas undergoing four steps: isothermal expansion at 500 K from 2.0 L to 6.0 L at initial pressure $3.0 \times 10^5 \text{ Pa}$; constant-volume cooling to 300 K; isothermal compression back to 2.0 L; and constant-volume heating back to 500 K. In the factory canteen, a kitchen assistant placed 50 g of ice at 0°C into a copper calorimeter of mass 100 g already holding 200 g of water at 20°C , and a physics intern was asked to predict the final temperature of the mixture.

Support Material Provided:

Refractive index of glass $n = 1.52$

Specific heat capacity: copper = $400 \text{ J kg}^{-1} \text{ K}^{-1}$, water = $4200 \text{ J kg}^{-1} \text{ K}^{-1}$

Specific latent heat of fusion of ice = $3.36 \times 10^5 \text{ J kg}^{-1}$

$\ln 3 = 1.099$

Tasks:

Support Dr. Ssali's team demonstration by calculating the angle of refraction and the critical angle for the glass used in the fibre, and drawing a labelled cross-sectional diagram of an optical fibre to explain how total internal reflection enables a light pulse to travel from Kampala to Gulu without escaping through the walls of the fibre. (08)

Assist Mr. Muwanga by sketching a fully labelled P-V diagram for the kiln's four-step thermodynamic cycle, calculating the work done by the gas during the Isothermal expansion at 500 K, and applying the first law of thermodynamics to determine the heat exchanged during each constant-volume process, clearly indicating whether heat is absorbed or released in each case. (08)

© Analyse the canteen calorimetry problem: show by calculation whether all 50 g of ice will melt when placed in the calorimeter, and determine the final equilibrium temperature of the mixture. (04),

SECTION D (FIELDS)

Answer one question from this section

Item 6(20marks)

During a dry afternoon physics practical at Greenhill Secondary School, a learner, Janet, rubbed a plastic comb with a woolen cloth and noticed it attracted small pieces of paper. She then used the same cloth to wipe the terminals of a resistor before setting up a meter bridge circuit to investigate how temperature affects resistance.

Janet placed the resistor X (nichrome wire) in the left gap and a standard resistor $S = 10.0 \, \Omega$ in the right gap. At 20°C , the balance point was at 45.0 cm from the left end. After heating resistor X, she recorded the following data:

Temperature ($^\circ\text{C}$)	Balance length l (cm)
20	45.0
40	42.3
60	39.8
80	37.5
100	35.4

TASK:

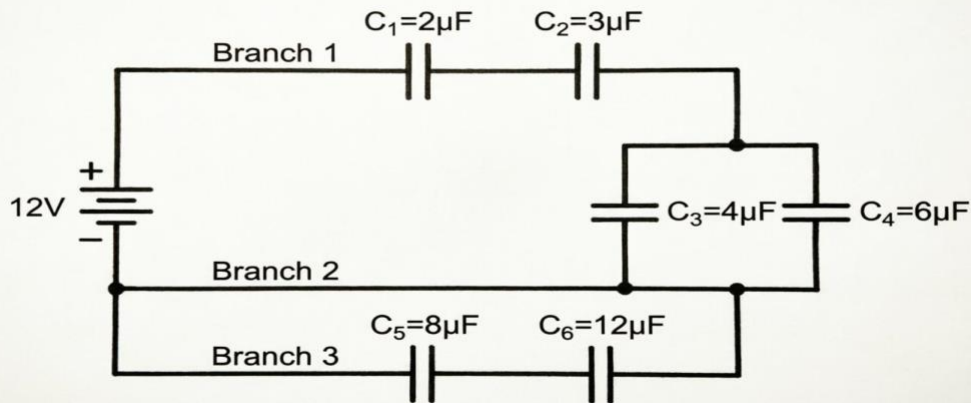
As a physics learner;

- Explain why the plastic comb attracted paper after rubbing but the metallic resistor terminals did not show the same effect, and state why Janet wiped the terminals before taking measurements. (4 marks)
- Using the meter bridge principle, calculate the resistance of resistor X at 20°C and at 100°C . (5 marks)
- Plot a graph of resistance of X (y-axis) against temperature (x-axis) and use it to determine the temperature coefficient of resistance (α) for the nichrome wire. (7 marks)
- Suggest TWO ways electrostatic charge buildup during the experiment could affect the meter bridge readings, and state ONE precaution Janet could take to minimize this interference. (4 marks)

Item 7(20marks)

A rural community health centre experiences frequent power grid fluctuations, causing sensitive medical refrigerators for vaccines to cycle off unexpectedly. To solve this, a local engineer designs a capacitor bank to act as a short-term voltage stabilizer that bridges micro-outages. The bank uses six capacitors connected to a 12V DC backup source as shown in the circuit diagram below

Circuit Diagram: Capacitor Bank Configuration



TASK:

As a physics learner;

- Explain TWO physical factors that determine the capacitance of each capacitor in the bank, and state why the engineer must carefully select the dielectric material given that the health centre lacks air conditioning and experiences high ambient temperatures. (4 marks)
- Calculate the effective capacitance of the entire capacitor bank configuration. (5 marks)
- Determine the total electrical energy stored in the bank when fully charged, and calculate the charge stored specifically on capacitor C_3 . (6 marks)
- During field testing, the engineer notices the bank loses charge faster than theoretical calculations predict. Suggest ONE practical reason for this energy loss in real-world capacitors, and state ONE critical safety precaution the technician must follow before servicing or rearranging the bank. (5 marks)